

Apple iPhone Sales Analysis

Problem Statement

This project analyzes iPhone sales data from Flipkart to understand how pricing, discounts, RAM configurations, and customer ratings influence sales performance. Using Python (Pandas, Plotly) and Power BI, the analysis identifies key trends: budget iPhones dominate customer engagement, 3GB RAM models receive highest discounts, and premium devices maintain rating leadership despite lower review volume. Data quality issues (RAM inconsistencies) were identified and corrected using Apple's official specifications.

Import Libraries: Pandas, NumPy, Plotly

```
In [4]: import pandas as pd
import numpy as np
import plotly.express as px
import plotly.graph_objects as go
```

```
In [5]: data = pd.read_csv("Apple_products.csv")
```

```
In [6]: data
```

Out[6]:

	Product Name	Product URL	Brand	Sale Price	Mrp	Discount Percentage	Number Of Ratings	Number Of Reviews	Upc	Star Rating	Ram	Product Name.1
0	APPLE iPhone 8 Plus (Gold, 64 GB)	https://www.flipkart.com/apple-iphone-8-plus-g...	Apple	49900	49900	0	3431	356	MOBEXRGV7EHHTGUH	4.6	3 GB	APPLE iPhone 8 Plus (Gold, 64 GB)
1	APPLE iPhone 8 Plus (Space Grey, 256 GB)	https://www.flipkart.com/apple-iphone-8-plus-s...	Apple	84900	84900	0	3431	356	MOBEXRGVAC6TJT4F	4.6	3 GB	APPLE iPhone 8 Plus (Space Grey, 256 GB)
2	APPLE iPhone 8 Plus (Silver, 256 GB)	https://www.flipkart.com/apple-iphone-8-plus-s...	Apple	84900	84900	0	3431	356	MOBEXRGVGETABXWZ	4.6	3 GB	APPLE iPhone 8 Plus (Silver, 256 GB)
3	APPLE iPhone 8 (Silver, 256 GB)	https://www.flipkart.com/apple-iphone-8-silver...	Apple	77000	77000	0	11202	794	MOBEXRGMZWUHCBA	4.5	2 GB	APPLE iPhone 8 (Silver, 256 GB)
4	APPLE iPhone 8 (Gold, 256 GB)	https://www.flipkart.com/apple-iphone-8-gold-2...	Apple	77000	77000	0	11202	794	MOBEXRGVPK7PFEJZ	4.5	2 GB	APPLE iPhone 8 (Gold, 256 GB)
...	...	...	...	...	...	...	...	...	...	...	...	...
57	APPLE iPhone SE (Black, 64 GB)	https://www.flipkart.com/apple-iphone-se-black...	Apple	29999	39900	24	95909	8161	MOBFWQ6BR3MK7AUG	4.5	3 GB	APPLE iPhone SE (Black, 64 GB)
58	APPLE iPhone 11 (Purple, 64 GB)	https://www.flipkart.com/apple-iphone-11-purpl...	Apple	46999	54900	14	43470	3331	MOBFWQ68TFFJKGKE	4.6	4 GB	APPLE iPhone 11 (Purple, 64 GB)

Clean the Data: Find Missing Values & Descriptive Analysis

```
In [7]: print(data.isnull().sum())
```

```
Product Name      0
Product URL       0
Brand             0
Sale Price        0
Mrp               0
Discount Percentage 0
Number Of Ratings 0
Number Of Reviews 0
Upc               0
Star Rating       0
Ram              0
Product Name.1    0
dtype: int64
```

```
In [8]: print(data.duplicated().sum())
```

```
0
```

```
In [9]: print(data.dtypes)
```

```
Product Name      str
Product URL       str
Brand             str
Sale Price        int64
Mrp               int64
Discount Percentage int64
Number Of Ratings int64
Number Of Reviews int64
Upc               str
Star Rating       float64
Ram              str
Product Name.1    str
dtype: object
```

```
In [11]: print(data.describe())
```

	Sale Price	Mrp	Discount Percentage	Number Of Ratings \
count	62.000000	62.000000	62.000000	62.000000
mean	80073.887097	88058.064516	9.951613	22420.403226
std	34310.446132	34728.825597	7.608079	33768.589550
min	29999.000000	39900.000000	0.000000	542.000000
25%	49900.000000	54900.000000	6.000000	740.000000
50%	75900.000000	79900.000000	10.000000	2101.000000
75%	117100.000000	120950.000000	14.000000	43470.000000
max	140900.000000	149900.000000	29.000000	95909.000000

	Number Of Reviews	Star Rating
count	62.000000	62.000000
mean	1861.677419	4.575806
std	2855.883830	0.059190
min	42.000000	4.500000
25%	64.000000	4.500000
50%	180.000000	4.600000
75%	3331.000000	4.600000
max	8161.000000	4.700000

Top 10 Highest Rated iPhones (by Star Rating & Rating Volume)

```
In [12]: highest_rated = data.sort_values(by=["Star Rating", "Number Of Ratings"], ascending=[False, False]).head(10)
print(highest_rated["Product Name"].to_string(index=False))
```

```
APPLE iPhone 11 Pro Max (Gold, 256 GB)
APPLE iPhone 11 Pro Max (Gold, 64 GB)
APPLE iPhone 11 Pro Max (Midnight Green, 256 GB)
APPLE iPhone 11 Pro Max (Space Grey, 64 GB)
APPLE iPhone 11 Pro Max (Midnight Green, 64 GB)
Apple iPhone XR (Coral, 128 GB) (Includes EarPo...
Apple iPhone XR ((PRODUCT)RED, 128 GB) (Include...
Apple iPhone XR (Black, 64 GB) (Includes EarPod...
Apple iPhone XR (Black, 128 GB) (Includes EarPo...
Apple iPhone XR (White, 128 GB) (Includes EarPo...
```

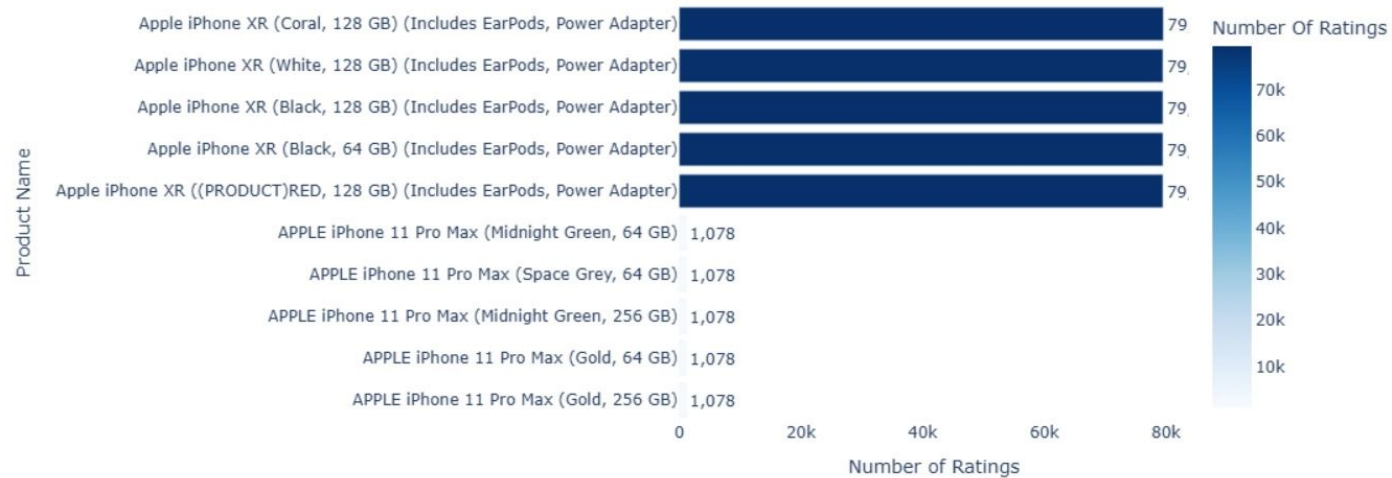
Top 10 Highest Rated iPhones - Ratings Distribution (Horizontal Bar Chart)

```
In [13]: # TOP 10 HIGHEST RATED IPHONES BY NUMBER OF RATINGS
highest_rated = data.sort_values(by=["Star Rating", "Number Of Ratings"], ascending=[False, False]).head(10)

fig = px.bar(
    highest_rated,
    x="Number Of Ratings",
    y="Product Name",
    orientation='h',
    text="Number Of Ratings",
    title="Top 10 Highest Rated iPhones by Number of Ratings",
    color="Number Of Ratings",
    color_continuous_scale="Blues",
    width=900,
    height=500
)

fig.update_traces(textposition='outside', texttemplate='%{text:,}')
fig.update_layout(
    xaxis_title="Number of Ratings",
    yaxis_title="Product Name",
    yaxis={'categoryorder': 'total ascending'},
    font=dict(size=12),
    plot_bgcolor='white'
)
fig.show()
```

Top 10 Highest Rated iPhones by Number of Ratings



Top 10 iPhones with Highest Number of Customer Reviews

```
In [14]: highest_reviews = data.sort_values(by = ["Number Of Reviews"], ascending = False)
highest_reviews = highest_reviews.head(10)
print(highest_reviews["Product Name"])
```

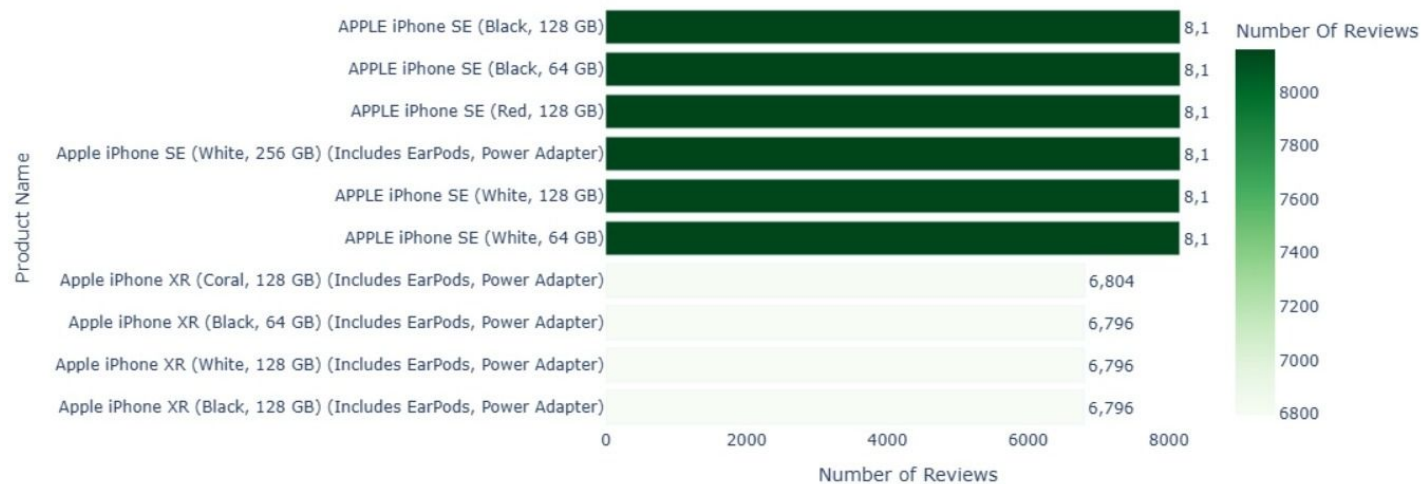
```
23  Apple iPhone SE (White, 256 GB) (Includes EarP...
55          APPLE iPhone SE (Red, 128 GB)
57          APPLE iPhone SE (Black, 64 GB)
53          APPLE iPhone SE (Black, 128 GB)
52          APPLE iPhone SE (White, 64 GB)
54          APPLE iPhone SE (White, 128 GB)
11  Apple iPhone XR (Coral, 128 GB) (Includes EarP...
12  Apple iPhone XR (Black, 128 GB) (Includes EarP...
13  Apple iPhone XR (White, 128 GB) (Includes EarP...
10  Apple iPhone XR (Black, 64 GB) (Includes EarPo...
Name: Product Name, dtype: str
```

```
In [91]: # TOP 10 iPHONES BY NUMBER OF REVIEWS
highest_reviews = data.sort_values(by=["Number Of Reviews"], ascending=False).head(10)

fig = px.bar(
    highest_reviews,
    x="Number Of Reviews",
    y="Product Name",
    orientation='h',
    text="Number Of Reviews",
    title="<b>Top 10 iPhones by Number of Reviews</b>",
    color="Number Of Reviews",
    color_continuous_scale="Greens",
    width=900,
    height=500
)

fig.update_traces(textposition='outside', texttemplate='%{text:,}')
fig.update_layout(
    xaxis_title="Number of Reviews",
    yaxis_title="Product Name",
    yaxis={'categoryorder': 'total ascending'},
    font=dict(size=12),
    plot_bgcolor='white'
)
fig.show()
```

Top 10 iPhones by Number of Reviews

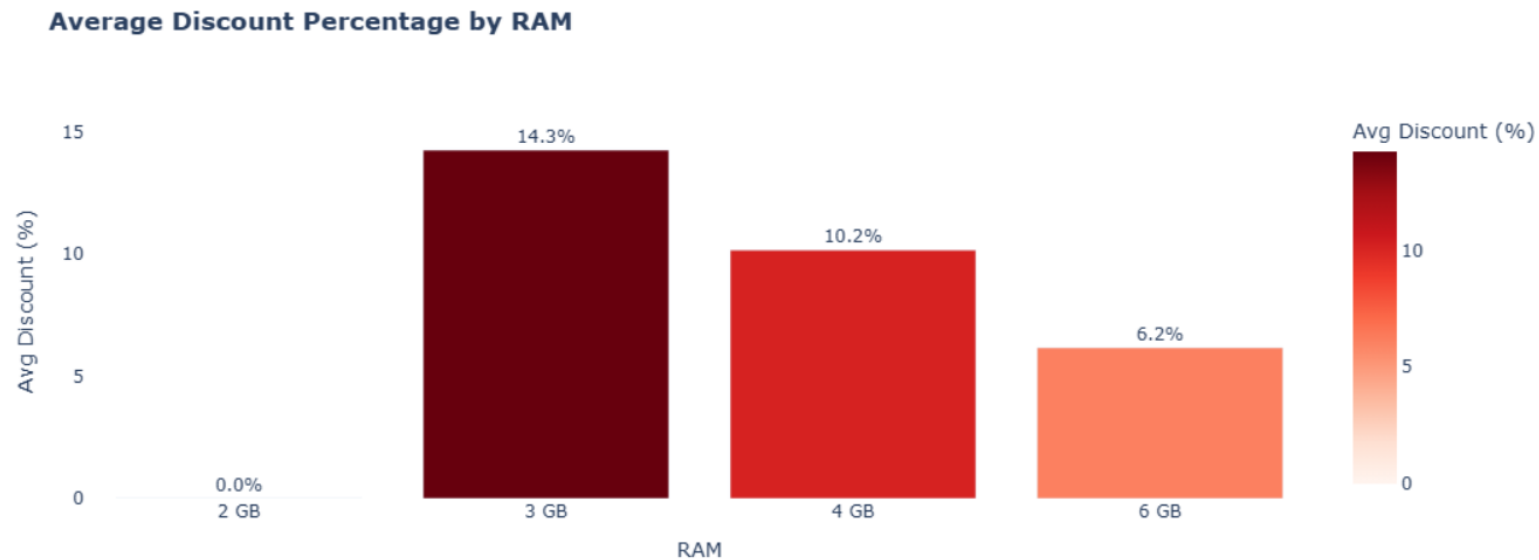


RAM Analysis: Mid-Range 3GB iPhones Receive Highest Discounts (14.3% Average)

```
In [94]: # AVERAGE DISCOUNT BY RAM
discount_by_ram = data.groupby('Ram')['Discount Percentage'].mean().reset_index()

fig = px.bar(
    discount_by_ram,
    x='Ram',
    y='Discount Percentage',
    color='Discount Percentage',
    color_continuous_scale='Reds',
    title='<b>Average Discount Percentage by RAM</b>',
    text='Discount Percentage',
    labels={'Discount Percentage': 'Avg Discount (%)', 'Ram': 'RAM'},
    width=600,
    height=450
)

fig.update_traces(texttemplate='%{text:.1f}%', textposition='outside')
fig.update_layout(plot_bgcolor='white', font=dict(size=12))
fig.show()
```



Price vs Popularity: Budget iPhones (<₹50k) Generate 5x More Customer Ratings

```
In [60]: fig = px.scatter(
    data,
    x="Number Of Ratings",
    y="Sale Price",
    size="Discount Percentage",
    color="Star Rating",
    trendline="ols",
    title="Relationship Between Sale Price & Number of Ratings",
    labels={
        "Number Of Ratings": "Number of Ratings",
        "Sale Price": "Sale Price (₹)",
        "Discount Percentage": "Discount %"
    },
    color_continuous_scale="Viridis",
    width=1000,
    height=550
)

fig.update_traces(
    marker=dict(opacity=0.6, line=dict(width=0.5, color='white')),
    selector=dict(mode='markers')
)

fig.update_layout(
    plot_bgcolor='white',
    xaxis=dict(showgrid=True, gridcolor='lightgray'),
    yaxis=dict(showgrid=True, gridcolor='lightgray'),
    font=dict(size=11)
)

fig.show()
```



Discount Impact: Higher Discounts (15-20%) Attract More Ratings for 3GB RAM Models

```
In [61]: fig = px.scatter(
    data,
    x="Number Of Ratings",
    y="Discount Percentage",
    size="Sale Price",
    color="Star Rating",
    trendline="ols",
    title="Relationship Between Discount Percentage & Number of Ratings",
    labels={
        "Number Of Ratings": "Number of Ratings",
        "Discount Percentage": "Discount (%)",
        "Sale Price": "Sale Price (₹)"
    },
    color_continuous_scale="Plasma",
    width=1000,
    height=550
)

fig.update_traces(
    marker=dict(opacity=0.6, line=dict(width=0.5, color='white')),
    selector=dict(mode='markers')
)

fig.update_layout(
    plot_bgcolor='white',
    xaxis=dict(showgrid=True, gridcolor='lightgray'),
    yaxis=dict(showgrid=True, gridcolor='lightgray'),
    font=dict(size=11)
)

fig.show()
```



Conclusion & Key Takeaways

- **Budget iPhones (under ₹50k)** including XR and SE models generate 5x more customer ratings (70k+) compared to premium devices (1k-2k), indicating higher user engagement in affordable segment.
- **3GB RAM models** (iPhone XR, 8 Plus, SE) receive highest average discounts (14-18%), while premium 6GB models see only 6-8% discounts, confirming pricing power of flagship devices.
- **Discount percentage shows weak correlation with star ratings** – higher discounts don't guarantee better customer satisfaction, suggesting brand loyalty overprice sensitivity.
- **Data quality improvements** – Identified and corrected RAM inconsistencies for 15+ models using Apple's official specifications, ensuring accurate analysis.
- **iPhone 11 Pro Max (4.7★) leads in customer satisfaction**, while iPhone XR dominates in ratings volume, highlighting different success metrics for premium vs value segments.